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Solid solution alloys of AlCoCrFeNiTi_x with excellent room-temperature mechanical properties

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Alloys with composition of AlCoCrFeNiTi_x (x : molar ratio; $x=0, 0.5, 1, 1.5$) were designed by using the strategy of equiatomic ratio and high entropy of mixing. The alloy system is composed mainly of body centered cubic solid solution and possesses excellent room-temperature compressive mechanical properties. Particularly for $\text{AlCoCrFeNiTi}_{0.5}$ alloy, the yield stress, fracture strength, and plastic strain are as high as 2.26 GPa, 3.14 GPa, and 23.3%, respectively, which are superior to most of the high-strength alloys such as bulk metallic glasses. © 2007 American Institute of Physics. [DOI: 10.1063/1.2734517]

The general strategy for developing alloys is to select one or two elements as principal components for primary properties and other minor elements for the acquisition of definite microstructure and properties, such as steels, TiAl intermetallics,¹ bulk metallic glasses,^{2–4} etc. Why not incorporate multiprincipal elements into alloy preparation is the anticipated formation of many intermetallic compounds and other complex microstructures, which would make alloys brittle and difficult in processing and analysis.⁵

In this letter, the strategy for developing alloys, e.g., equiatomic ratio and high entropy of mixing,^{6–9} was applied to the design of multicomponent AlCoCrFeNiTi_x alloys. Distinct from the traditional alloys, the alloy system contains more than five elements as its major constituent. Here, the effect of the alloying of Ti with quinary AlCoCrFeNi equiatomic ratio alloy on the microstructure and mechanical properties was studied.

Alloy ingots with nominal composition of AlCoCrFeNiTi_x (x : molar ratio; $x=0, 0.5, 1$, and 1.5 , denoted by Ti_0 , $\text{Ti}_{0.5}$, Ti_1 , and $\text{Ti}_{1.5}$, respectively) were prepared by arc melting mixtures of pure metals in a Ti-gettered high-purity argon atmosphere. The ingots were then remelted under high vacuum (10^{-3} Pa) and injection cast into a water-cooled copper mould to obtain cylindrical rods of $\varnothing 5 \times 70$ mm. The microstructure of as-cast samples was characterized by x-ray diffraction (XRD). The morphology of alloys was examined with a LEO-1450 scanning electron microscope (SEM). Compressive tests were carried out on samples of $\varnothing 5 \times 10$ mm with MTS 809 materials testing machine at room temperature with a strain rate of $1 \times 10^{-4} \text{ s}^{-1}$.

Figure 1 shows the XRD patterns of the as-cast samples. Interestingly, all the four alloys are composed mainly of bcc solid solution. Ti_0 alloy exhibits only single bcc phase reflections. While for the other three alloys, another bcc phase precipitates and becomes the main phase. The reflections of the main bcc phase gradually shift to the left with the increase of Ti addition, which indicates that the lattice parameter of bcc phase has a slight increment as more Ti with large atomic radius solves into bcc solid solution. For $\text{Ti}_{1.5}$ alloy, besides bcc reflections, there are reflections of Laves phase which has lattice parameters of $a=0.4765$ nm and $c=0.7723$ nm and can be identified as Fe_2Ti type.

SEM images of the alloys are shown in Fig. 2. The morphology of the four alloys is different from each other. Ti_0 alloy shows morphology of single phase, which is consistent with the XRD result. The other three alloys contain multiphase and exhibit typically cast dendrite structure, with the dendrite size and morphology similar to each other. The dendrites are round and irregular with an average primary arm width of 4–8 μm . In both $\text{Ti}_{0.5}$ and Ti_1 alloys, eutectic can be observed in the interdendrite region.

According to the Gibbs phase rule, $F=C-P+1$ (F : degree of freedom, C : number of components, and P : number of phase), the maximum number of equilibrium phases in a C component system at constant pressure is $P=C+1$. However, in this research, simple solid solution phases such as bcc are prior to intermetallic phases to form in AlCoCrFeNiTi_x alloys, and the total number of phases is well below the maximum equilibrium number allowed by the Gibbs phase rule.

The prior formation of simple solid solutions to intermetallic compounds in this alloy system is attributed to the effect of high entropy of mixing.¹⁰ Following Boltzmann's hypothesis, the entropy of mixing of a n -element regular solution alloys is as follows:

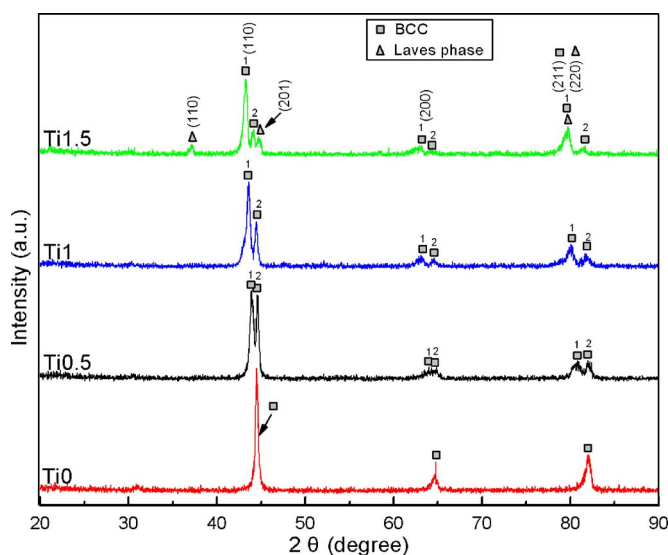


FIG. 1. XRD patterns of as-cast Ti_0 , $\text{Ti}_{0.5}$, Ti_1 , and $\text{Ti}_{1.5}$ alloys.

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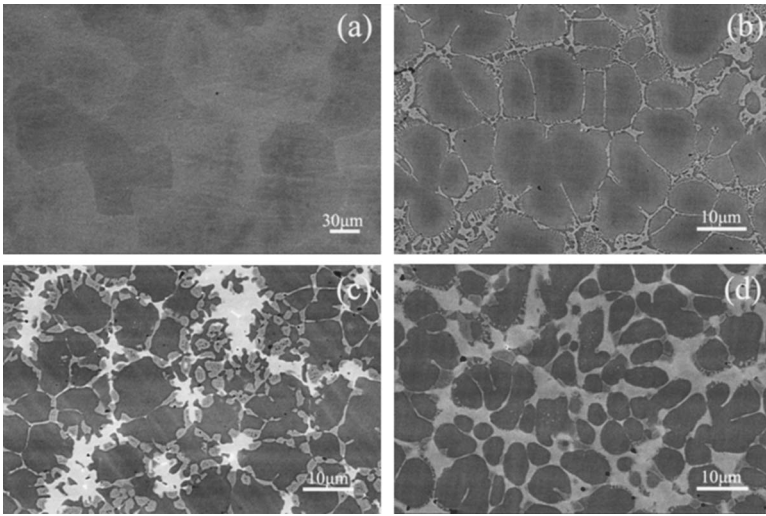


FIG. 2. SEM backscattered electron images of as-cast (a) Ti₀, (b) Ti_{0.5}, (c) Ti₁, and (d) Ti_{1.5} alloys.

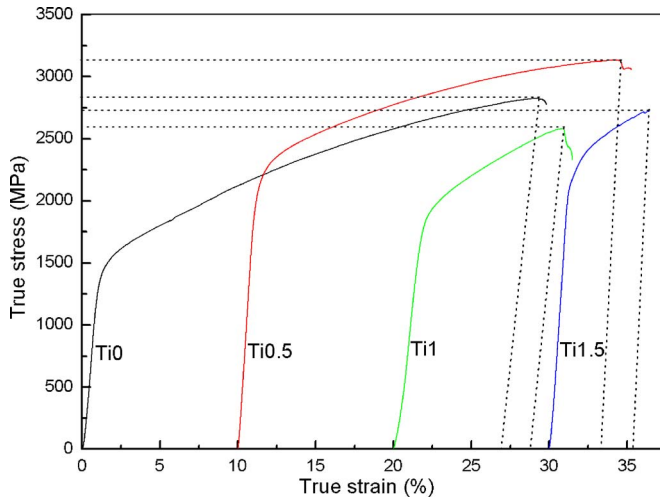


FIG. 3. Compressive true stress-strain curves of Ti₀, Ti_{0.5}, Ti₁, and Ti_{1.5} alloy rods with diameter of 5 mm.

$$S_{\text{mix}} = -R \sum_{i=1}^n c_i \ln c_i, \quad (1)$$

where $\sum_{i=1}^n C_i = 1$, C_i is mole percent of component, and R is gas constant. The entropy of mixing reaches maximum when the alloy is of equiatomic ratio, like Ti₁ alloy (AlCoCr-FeNiTi).

As following is the formula of Gibbs free energy:

$$G_{\text{mix}} = H_{\text{mix}} - TS_{\text{mix}}, \quad (2)$$

where H_{mix} is the enthalpy of mixing and T is the absolute temperature.

According to Eq. (2), the high entropy of mixing can significantly lower the free energy, thus lowering the tendency to order and segregate, and consequently making random solid solution more easily form and more stable than intermetallics or other ordered phases during solidification in alloys with multiprincipal elements, especially at high temperature.^{11–13}

Figure 3 shows the room-temperature compressive true stress-strain curves of the alloy system. All the alloys exhibit fairly high yield stress, fracture strength, plastic deformation, and work hardening capacity. The detailed mechanical properties for the alloy system and some of the other high-

TABLE I. Room-temperature compressive mechanical properties for Ti₀, Ti_{0.5}, Ti₁, and Ti_{1.5} rod samples and some of other high-strength bulk alloys (BMG: bulk metallic glasses; LOS: long-period order structure). Young's modulus (E), yield stress (σ_y), fracture strength (σ_{max}), elastic strain (ϵ_e), and plastic strain (ϵ_p).

Alloy	Microstructure	E (GPa)	σ_y (GPa)	σ_{max} (GPa)	ϵ_e (%)	ϵ_p (%)	Refs.
Ti ₀	bcc solid solution	127.0	1.50	2.83	1.0	26.9	This work
Ti _{0.5}	bcc solid solution	177.7	2.26	3.14	1.1	23.3	This work
Ti ₁	bcc solid solution	90.1	1.86	2.58	1.7	8.8	This work
Ti _{1.5}	bcc+Laves	159.8	2.22	2.72	1.3	5.3	This work
Mg ₈₁ Cu _{9.3} Y _{4.7} Zn ₅	BMG+ <i>in situ</i> LOS	19.2	0.55	1.16	2.9	18.5	14
Ti ₆₀ Cu ₁₄ Ni ₁₂ Sn ₄ Nb ₁₀	Nanostructure+ <i>in situ</i> BCC dendrite	66.1	1.31	2.40	2.2	14.5	15
Zr _{56.2} Ti _{13.8} Nb ₅ Cu _{6.9} Ni _{5.6} Be _{12.5}	BMG+ <i>in situ</i> BCC dendrite	110.0	1.30	1.70	1.2	6.8	16
Cu ₄₈ Zr ₄₈ Al ₄	BMG+CuZr phase	102.8	1.20	1.88	2.2	5.3	17
Ti ₄₀ Zr ₂₅ Ni ₃ Cu ₁₂ Be ₂₀	BMG	90.0	1.68	1.78	1.8	3.2	18
Zr _{41.2} Ti _{13.8} Cu _{12.5} Ni ₁₀ Be _{22.5}	BMG	96.3	1.64	1.76	1.8	0.3	19
Co ₄₃ Fe ₂₀ Ta _{5.5} B _{31.5}	BMG	268.0	5.19	5.19	2.1	...	20

strength bulk alloys are listed in Table I. $\text{Ti}_{0.5}$ alloy exhibits yield stress, fracture strength, and plastic strain of 2.26 GPa, 3.14 GPa, and 23.3%, respectively. It is apparent that these properties are even superior to most of the high-strength bulk alloys reported, e.g., bulk metallic glasses.

The excellent comprehensive mechanical properties of $\text{Ti}_{0.5}$ alloy are attributed to solid solution strengthening mechanism of Ti atoms. For Ti_0 alloy, the five components Al, Co, Cr, Fe, and Ni are equiatomic ratio and have the same probability to occupy the lattice sites to form bcc solid solution. As the Ti with larger atomic radius occupies the lattice sites, the lattice distortion energy will increase significantly and the effect of solid solution strengthening is enhanced, thus the strength greatly increases, with the plasticity slightly decreasing.

In summary, by applying the strategy of equiatomic ratio and high entropy of mixing, AlCoCrFeNiTi_x alloys are designed and studied on its microstructure and mechanical properties. The alloy system is composed mainly of bcc solid solution and possesses excellent room-temperature compressive mechanical properties. Among them, $\text{Ti}_{0.5}$ alloy, whose yield stress, fracture strength, and plastic strain are as high as 2.26 GPa, 3.14 GPa, and 23.3% respectively, has the best comprehensive mechanical properties even superior to most of the high-strength alloys such as bulk metallic glasses.

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